



PTH6121: Musculoskeletal Conditions and Management

Fall 2026

Fridays 9:30-12:00

In Person

Course Information:

Semester: Fall 2026

Credit Hours: 3

Semester Clock Hours: 38

Weekly Clock Hours: 2.5

Instructor:

Name: Dr. Timothy Duffy

Email: tduffy1@plymouth.edu

Office hours: I am available before and after class and will offer virtual office hours by appointment via Microsoft Teams.

Pre Requisites: PTH 6110, PTH 6111, PTH 6115, PTH 6112, PTH 6116.

Co Requisites: PTH 6120, PTH 6122, PTH 6124, PTH 6125, PTH 6127.

Course Description

Musculoskeletal conditions and management covers common muscular, skeletal and connective tissue pathology, medical and surgical management across the lifespan with implications for physical therapy practice. Students will learn the basics of radiographic and MRI imaging technologies and develop an understanding of the use of such technologies in the diagnosis and management of patients with musculoskeletal conditions. This course will provide instruction on the basic science behind each condition and students will present on various orthopedic interventions encountered by physical therapists.

Student Learning Outcomes (course objectives)

1. Describe the components and integration of the musculoskeletal system (muscle, skeletal, connective tissue). (7A)
2. Identify on a radiograph the components and relationships of the musculoskeletal system. (7A, 7C)
3. Identify the results of imaging on the practice of physical therapy with respect to the musculoskeletal system. (7A, 7D17)
4. Identify the need for various imaging procedures in the examination of the patient/client with musculoskeletal conditions. (7A, 7D20)
5. Identify the musculoskeletal conditions amenable to physical therapy interventions and describe the underlying pathophysiology. (7A, 7D16)

6. Discuss the impact of mobility on the rate of healing in bone and other tissues. (7A)
7. Describe the pathological conditions of the musculoskeletal system commonly encountered by physical therapists. (7C)
8. Describe the medical and surgical management of pathological conditions of the musculoskeletal system commonly seen by physical therapists. (7C)
9. Describe the contributions of health care professionals other than physical therapists in the management of pathological conditions of the musculoskeletal system commonly seen by physical therapists. (7D16)
10. Identify the general categories, mechanisms of action and risk-benefit ratio for the pharmacological agents routinely prescribed for relief of pain in the musculoskeletal system. (7D17)

Required Materials:

There are no required readings for this course. Supplemental reading materials are outlined in the syllabus. The first half of this course discusses basic principles of musculoskeletal conditions and it is recommended to supplement lecture material with readings in areas you are less familiar with. A subscription to PhysioU is required for this course and this subscription contains every reference listed in the schedule which is highlighted in purple.

Recommended Materials:

These texts are **not required** but can be a helpful supplement to lecture and accessphysio materials. The Salter text is a classic staple in many graduate level healthcare programs and can therefore be purchased in used condition for less than \$20.

1. Manske, R, Magee, D, Zachazewski, J, Quillen, W. *Pathology and Intervention in Musculoskeletal Rehabilitation*, 2e. Saunders; 2016
2. Salter, R. *Textbook of Disorders and Injuries of the Musculoskeletal System*, 3e. Lippincott Williams & Wilkins; 1999

Description of teaching methods and learning experiences

Lectures, group discussions, group presentations

Assignments and Activities

Exam 1.....	25%
Exam 2.....	25%
Weekly Quizzes.....	10%
Group Presentations...	10%
Cumulative Final.....	30%

Weekly Quizzes

There will be 12 quizzes at the beginning of each lecture that does not have a scheduled exam. These will be 5-6 questions in length, A-D multiple choice and will test the material covered from the previous. Quiz 1 will test the material on embryology.

Exams

50 question multiple choice format, the final will have both new and cumulative content

Group Presentations

Students will present various orthopedic interventions encountered by physical therapists. The format will consist of:

10 groups of 2-3 students
15 minute presentations

Include the following information:

- 1) Indications of the intervention (why is it performed?)
- 2) The patient population likely undergoing the intervention (what is the pre-operative diagnosis? If there is a pathological process or mechanism of injury describe it)
- 3) Describe the procedure (type of surgical approach, estimated length of procedure, describe the involved anatomy and how it is changed)
- 4) Describe the prognosis (what can a patient reasonably expect to do following surgery?)
- 5) Discuss the post-operative rehab pearls (what structures do you need protect initially? (Protocols are an excellent resource for this)

Presentation Schedule:

Reverse total shoulder arthroplasty (rTSA)	Group 1
Rotationplasty	Group 2
Anterior cervical discectomy and fusion (ACDF)	Group 3
CMC Arthroplasty	Group 4
Ulnar Collateral Ligament Reconstruction	Group 5
Latarjet Procedure	Group 6
Posterior lumbar interbody fusion (PLIF)	Group 7
Periacetabular Osteotomy	Group 8
Quadriceps Tendon Repair	Group 9
Triple arthrodesis	Group 10

Groups have been assigned on Canvas. Go to People then click on Group Presentations

Student Evaluation/Grading Scale

A	94.0 – 100
A-	90.0 – 93.9
B+	87.0 – 89.9
B	83.0 – 86.9
B-	80.0 – 82.9
C+	77.0 – 79.9
C	73.0 – 76.9
F	below 73.0

Students are reminded that continuation in the program necessitates achieving an overall grade point average of 3.0 (B) or better. Grades below 2.0 (C) are not acceptable or counted toward the DPT degree. No more than 10 credits below 3.0 may be counted toward the DPT degree (grades of B-, C+, or C). Make-up examinations will be given only in specific cases of absence (documented absences). Circumstances that may warrant an absence should be discussed with the instructor before the scheduled time of the exam. Failure to notify the instructor as stated may result in a zero for the exam.

Professional Conduct: Students are reminded of the program expectation that they follow Professional Conduct, see the Student Handbook, Professional Conduct and Reasonable Professional Expectations for the process of handling violations, remediation and consequences.

Cell Phones: Cell phone use (including texting) is prohibited in class. Also, please respect your classmates by turning your phone to vibrate prior to class start time. Do not turn phones off as the University uses texts to relate emergency information.

Graduate Academic Policies: The DPT Program is a selective admissions cohort-structured program. Some Graduate Academic Policies may not be relevant to the program. Students should refer to the Doctor of Physical Therapy Program Handbook for specific information about Scheduling, Capstones, and Internships. The sections on Grading System and Academic Standing Policy do apply to DPT students:

<https://coursecatalog.plymouth.edu/university-policies-procedures/graduate-academic-policies/>

Fair Grading Policy: Fair and equitable grading reflects values to which all members of the Plymouth State community commit themselves. Grades are used to assess the relative extent to which students achieve course objectives in all for-credit courses at PSU. Academic freedom allows instructors to determine course objectives within the bounds of established curricula, and the means by which a student's mastery of those objectives will be evaluated; and evaluate the quality of work on individual exams or assignments. Information can be

found at <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Fair Grading.

Student Support: Student Support Foundation (SSF) provides short-term emergency financial assistance and long-term student support. For more information, see:

<https://campus.plymouth.edu/student-support-foundation/>

Student Support Foundation also runs a food pantry, located in Belknap Hall. To learn more about SSF or access the food pantry, either via open hours or a private appointment, contact the SSF advisor, at psu-ssf@plymouth.edu.

Academic Dishonesty: Cheating and plagiarism are very serious forms of academic dishonesty. Any use of unauthorized assistance on exams, papers, homework assignments, or other course work constitutes cheating. Knowingly providing assistance during an exam or allowing other students to copy one's work is also a serious form of academic dishonesty. Plagiarism consists of submitting written work that has been developed wholly or partially by someone else. Submitting written work in which the ideas of others have been duplicated or even paraphrased without proper reference to the author is also a form of plagiarism. Also considered plagiarism is the acquisition of term papers or other assignments from another source and subsequent presentation of these materials as the student's own work. In addition, students may not use papers in more than one course without the permission of both instructors.

Any student suspected of academic dishonesty will be reported to the Program Director immediately. A full description of academic integrity, violations and procedures when a violation to academic integrity in a course is suspected can be found at:

<https://coursecatalog.plymouth.edu/university-policies-procedures/> under Academic Integrity.

Attendance Policy: Attendance is expected to all class sessions. Students are ultimately responsible for their attendance, however faculty will assess students' Professional Expectations based, in part, on their attendance (See Student Evaluation/Grading). Students who miss class for any reason are responsible for the information and work missed. Students should speak with the professor before missing the scheduled class when able.

Campus Accessibility Policy: Plymouth State University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you should immediately contact Campus Accessibility Services (CAS) in Speare 210 (535-3300) to determine whether you are eligible for such accommodations. Academic accommodations will only be considered for students who have registered with CAS. If you have a Letter of Accommodation for this course from CAS, please provide the instructor with that information privately so that you

and the instructor can review those accommodations:

<https://campus.plymouth.edu/accessibility-services/>

LibGuide for Physical Therapy:

The Libguide for Physical Therapy is an information sharing platform created through a collaboration between the library and the DPT faculty that presents relevant resources to support students in the Physical Therapy program. Examples of information you will find include tutorials to better understand citing sources in APA and AMA style, resources on how to avoid plagiarism, quick access to relevant databases and interlibrary loan as well as a contact link to the programs' assigned librarian. To view the DPT Program Libguide go to <https://library.plymouth.edu/physical-therapy>.

Diversity and Inclusivity Statement:

We consider this classroom to be a place where you will be treated with respect, and we welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. See this web page for the Plymouth State University IDEA Center Inclusion, Diversity, Equity, and Access. <https://www.plymouth.edu/idea-center>

2026 Tentative Schedule

Week	Date	Topic	Quiz	Group Presentation	Supplemental Readings
1	9/4	Introduction, MSK development, principles of tissue healing	1	None	<i>Dutton</i> Chapter 2: Tissue Behavior, Injury, Healing, and Treatment <i>Salter</i> Ch. 2: Normal Structure and Function of the Musculoskeletal System
2	9/11	Bone healing, soft tissue injury classification and healing	2	1. Reverse total shoulder arthroplasty (rTSA)	<i>Dutton</i> Chapter 2: Tissue Behavior, Injury, Healing, and Treatment <i>Salter</i> Ch. 3: Reactions of Musculoskeletal Tissues to Disorders and Injuries
3	9/18	Peripheral nerve injury, Radiology ,Osteoarthritis, Generalized diseases of bone	3	2. Rotationplasty	<i>Mitra</i> Ch. 12: Peripheral nerve injuries: mononeuropathy <i>Kemp</i> Ch. 19. Pathology of the Bones and joints <i>Salter</i> Ch. 9 Generalized and Disseminated Disorders of Bone
4	9/25	Fractures	4	3. Anterior cervical discectomy and fusion (ACDF)	<i>Dutton</i> Chapter 2: Tissue Behavior, Injury, Healing, and Treatment <i>Salter</i> Ch. 15 Fractures and Joint Injuries - General Features
5	10/2	Inflammatory diseases of bone	5	4. CMC Arthroplasty	<i>Kemp</i> Ch. 19. Pathology of the Bones and Joints <i>Salter</i> Ch. 10 Inflammatory disorders of bones and joints
6	10/9	Generalized congenital deformities Epiphyseal plate injuries	---	EXAM ONE (weeks 1-5, <i>no group assignment</i>)	Hoogenboom Ch. 30: Considerations for the pediatric patient <i>Salter</i> Ch. 8 Congenital Abnormalities <i>Salter</i> Ch. 13 Disorders of Epiphysis and Epiphyseal Growth
7	10/16	Hand, Wrist, TOS	6	5. Ulnar Collateral Ligament Reconstruction	Variable material from accessphysio <i>Salter</i> Ch. 17 Specific Fractures and Joint Injuries in Adults <i>Magee</i> Chapters. 9-10
Dutton M. eds. <i>Dutton's Orthopaedic Examination, Evaluation, and Intervention</i>, 6e. McGraw Hill Education; 2023					
Kemp WL, Burns DK, Brown TG. eds. <i>Pathology: The Big Picture</i>. The McGraw-Hill Companies; 2008					
Manske, R, Magee, D, Zachazewski, J, Quillen, W. <i>Pathology and Intervention in Musculoskeletal Rehabilitation</i> , 2e. Saunders; 2016					
Mitra R. eds. <i>Principles of Rehabilitation Medicine</i>. McGraw-Hill Education; 2019.					
Salter, R. <i>Textbook of Disorders and Injuries of the Musculoskeletal System</i> , 3e. Lippincott Williams & Wilkins; 1999					
Hoogenboom BJ, Voight ML, Prentice WE. eds. <i>Musculoskeletal Interventions: Techniques for Therapeutic Exercise</i>, 4e. McGraw Hill; 2021					

Supplemental Readings (texts highlighted in purple are available on PhysioU)

Week	Date	Topic	Quiz	Group Presentation	Supplemental Readings
8	10/23	Elbow, Shoulder	7	6. Latarjet Procedure	Variable material from accessphysio <i>Salter</i> Ch. 17 Specific Fractures and Joint Injuries in Adults <i>Magee</i> Chapters. 5-8
9	10/30	Spine	8	7. Posterior lumbar interbody fusion (PLIF)	Variable material from accessphysio <i>Salter</i> Ch. 17 Specific Fractures and Joint Injuries in Adults <i>Magee</i> Chapters. 3, 13-16
10	11/6	Hip	9	8. Periacetabular Osteotomy	Variable material from accessphysio <i>Salter</i> Ch. 17 Specific Fractures and Joint Injuries in Adults <i>Magee</i> Chapter 18
11	11/13	Musculoskeletal Pharmacology	---	EXAM TWO (weeks 6-10, <i>no group assignment</i>)	Variable material, see lecture for supplemental readings
12	11/20	Knee	10	9. Quadriceps Tendon Repair	Variable material from accessphysio <i>Salter</i> Ch. 17 Specific Fractures and Joint Injuries in Adults <i>Magee</i> Chapters 20-22
-----	11/27	THANKSGIVING HOLIDAY NO CLASS			
13	12/4	Ankle	11	10. Triple arthrodesis	Variable material from accessphysio <i>Salter</i> Ch. 17 Specific Fractures and Joint Injuries in Adults <i>Magee</i> Chapters. 5-8
14	12/11	Content we did not have time to cover in the first 13 weeks plus: mini-lab and final exam review	12	None	None
15	12/18	Finals Week		FINAL EXAM (weeks 11-13 AND weeks 1-10)	
Dutton M. eds. <i>Dutton's Orthopaedic Examination, Evaluation, and Intervention</i>, 6e. McGraw Hill Education; 2023					
Kemp WL, Burns DK, Brown TG. eds. <i>Pathology: The Big Picture</i>. The McGraw-Hill Companies; 2008					
Manske, R, Magee, D, Zachazewski, J, Quillen, W. <i>Pathology and Intervention in Musculoskeletal Rehabilitation</i> , 2e. Saunders; 2016					
Mitra R. eds. <i>Principles of Rehabilitation Medicine</i>. McGraw-Hill Education; 2019.					
Salter, R. <i>Textbook of Disorders and Injuries of the Musculoskeletal System</i> , 3e. Lippincott Williams & Wilkins; 1999					
Hoogenboom BJ, Voight ML, Prentice WE. eds. <i>Musculoskeletal Interventions: Techniques for Therapeutic Exercise</i>, 4e. McGraw Hill; 2021					

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